

CS5286: ALGORITHMS AND TECHNIQUES FOR WEB SEARCHING

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Algorithms and Techniques for Web Searching

Subject Code

CS - Computer Science

Course Number

5286

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course teaches principles, tools and techniques for Internet information retrieval. Students will be able to develop automated access software to webpages, to analyze their link structures, to index them according to their contents, to rank them with respect to specific queries, to apply information retrieval tools, computational linguistics and hyperlink graph structural understandings to improve the searching results. Students will also be given chances to create their own innovative ideas for web searching in the group project.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Create an automated software agent for web searching.			x	
2	Analyze webpage through hyperlink structures.		x		
3	Index webpages by different document models for different purposes, and remember those models.		x		
4	Explain answering queries in decreasing orders of relevance, and evaluate the relevance.		x		
5	Creatively apply web search tools.				x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Concepts, theory and methodologies.	1, 2, 3, 4	2 hours/week
2	Tutorials	Exercises and discussions.	2, 3, 4, 5	1 hour/week
3	In-class exercise	Students are asked to practice what they learned in lectures to solve some related problems. Solutions by volunteers or selected students will be discussed in class.	2, 3, 4	

4	Programming assignment	The students are asked to program different search engines based on algorithms, techniques and engineering tricks learnt in the course. The assignments give students an opportunity to demonstrate the understanding of course materials, while exploiting popular APIs for web searching.	1	
5	Project	Two different types of projects are designed that allow students to choose from: either an essay based one, or a programming based one. Software-based assignment gives students an opportunity to create innovative applications for web searching. The assignment will be documented in a report. The essay writing project will be about a study on a specific aspect of web search. This activity supports Course ILO #5.	5	
6	Examination	Students will be tested on their overall understanding of the topics covered in CILO 1, 2, 3, 4.	1, 2, 3, 4	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- for nil entry)	Allow Use of GenAI?
1	Assignment	1, 2, 3	30	-	Yes
2	Project	3, 4, 5	20	Can be group project of 2-3 members	Yes

Continuous Assessment (%)

50

Examination (%)

50

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

A formative assessment will be made on the students' ability to apply tools and knowledge to different situation. The equal weighting of coursework and examination assessment is due to the emphasis on both the practicality and the theory of market design models.

Assessment Rubrics (AR)

Assessment Task

Assignment (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability to implement and assess the effectiveness of different algorithms and techniques.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability and creativity in applying appropriate algorithms and techniques for real-world applications.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The extent to which the students can understand the algorithms and techniques, apply them appropriately for different applications, and evaluate their performances.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Assignment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability to implement and assess the effectiveness of different algorithms and techniques.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability and creativity in applying appropriate algorithms and techniques for real-world applications.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The extent to which the students can understand the algorithms and techniques, apply them appropriately for different applications, and evaluate their performances.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal level

Part III Other Information

Keyword Syllabus

Search engines; web crawlers; web structure mining; page rank; micro communities; web content mining; inverted file index; vector space model; web advertisement; relevance feedback; near-duplicate search; probabilistic information retrieval; language models; topic modelling; multimedia search; deep learning for information retrieval; performance evaluation.

Syllabus:

1. Build an automated software agent for web searching: data preprocessing, indexing, search algorithms and techniques.
2. Analyze webpage through hyperlink structures: Web structure mining, page rank, micro communities.
3. Index webpages by different document models for different purposes: inverted indexing, Boolean model, vector space mode, near-duplicate search, probabilistic information retrieval, topic modelling, deep learning.
4. Answer queries in decreasing orders of relevance: similarity, query expansion, relevance feedback, performance evaluation, user behaviour analysis.
5. Creatively apply web search tools: Query log mining, distributed information retrieval, document clustering and classification.

Reading List**Compulsory Readings**

Title	
1	C.D. Manning, P. Raghavan, H. Schutze, "Introduction to Information Retrieval", Cambridge University Press, 2008, New York.

Additional Readings

Title	
1	Ricardo Baeza-Yates and Berthier Ribeiro-Neto, "Modern Information Retrieval", Addison-Wesley, 1999.
2	A collection of readings from journal literature and conference proceedings will be used